

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.2: Reflection and Congruence — Teacher Reference (Pre-filled)

SUMMARY TABLE: CORE MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.2: Reflection and Congruence
Driving Question	DRIVING QUESTION: Why does a matatu's rear-view mirror show an image that is the same size and shape as the real object behind it — yet appears flipped — and how can we use this to prove that the original and its mirror image are always congruent?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Symmetry All Around Us — Anchoring the Matatu Mirror Phenomenon & Lines of Symmetry	Learners observed that a matatu conductor's mirror on the Nairobi–Thika highway produces an image of a lorry that appears the same size as the real lorry yet is laterally flipped (the lorry's right headlight appears on the left in the mirror). Through paper-folding, learners also observed that plane figures can have different numbers of lines of symmetry — a square has 4, a rectangle has 2, an equilateral triangle has 3, the letter 'A' has 1, and the letter 'F' has 0.	A line of symmetry is a line that divides a plane figure into two congruent halves, each of which is a mirror image of the other. The mirror held by the matatu conductor acts as a mirror line (line of reflection). Figures with lines of symmetry include most regular polygons and many Kenyan cultural objects (Maasai beadwork, the Kenya flag), while asymmetric figures like the letter 'F' have none. The fold line in a paper cut-out is equivalent to the mirror line in a reflection.	The fact that the matatu mirror produces an image that is the same size as the real lorry — even though it appears flipped — is the first clue that reflection preserves shape and size (congruence). When we folded a rectangle along its line of symmetry, the two halves matched perfectly: same shape, same size, but one was a 'flip' of the other. This is exactly what the mirror does to the lorry. The puzzle of why the flipped image is still the same size and shape as the original will be the central question we investigate throughout this unit.
Lesson 2 Reflection All Around Us — Locating Lines of Symmetry in Plane Figures	Learners observed that placing a plane mirror along certain lines on a maize leaf, bottle cap, and paper shapes produces a reflection that completes the original figure perfectly — and that different shapes (equilateral triangle, rectangle, rhombus, regular hexagon) have different numbers of such lines. They also observed symmetry in Maasai beadwork and kikoi fabric designs.	A line of symmetry divides a plane figure into two congruent halves, each of which is the exact mirror image of the other. An equilateral triangle has 3 lines, a rectangle has 2, a rhombus has 2 (along its diagonals), and a regular hexagon has 6. Some figures have one line, some have many, and some have none.	The matatu mirror shows a perfect image of the lorry because the mirror surface acts as a line of symmetry — it divides the combined 'lorry + image' system into two congruent halves. This is WHY the image looks exactly like the lorry: the mirror line guarantees that one half (the image) is a perfect, congruent copy of the other half (the real lorry).
Lesson 3 Reflection on the	Learners plotted the vertices of a lorry-shaped quadrilateral and its reflected	Each mirror line produces a specific, predictable coordinate rule: reflection in the	Because coordinate reflection rules preserve every distance between vertices,

Cartesian Plane — Plotting Object and Image Coordinates	images across four mirror lines (x-axis, y-axis, $y = x$, $y = -x$) on Cartesian grids, observing that each image vertex is exactly as far from the mirror line as the original vertex, that the joining line is perpendicular to the mirror line, and that all side lengths of the image equal the corresponding side lengths of the object.	y-axis maps $(a, b) \rightarrow (-a, b)$; in the x-axis maps $(a, b) \rightarrow (a, -b)$; in $y = x$ maps $(a, b) \rightarrow (b, a)$; in $y = -x$ maps $(a, b) \rightarrow (-b, -a)$. These rules preserve all distances between points, meaning the image is always the same size and shape as the object.	the lorry image in the matatu rear-view mirror must be exactly the same size and shape as the real lorry behind the vehicle — the coordinates change in a systematic (flipped) way, but no length is stretched or shrunk. This is the mathematical reason the conductor can use the mirror to judge the lorry's true size and distance, and it is the first precise evidence that the object and its mirror image may be congruent.
Lesson 4 Properties of Reflection — Discovering the Four Rules of the Mirror Line	Using a plane mirror placed on a drawn mirror line, learners observed that the image of the matatu arrow appeared the same size as the object, appeared to be the same distance behind the mirror line as the object was in front of it, and appeared laterally flipped. Measuring the distances from object points and image points to the mirror line confirmed equal distances, and a set-square check confirmed that the connecting segments met the mirror line at 90° . Folding tracing paper along the mirror line showed that the object and image coincide exactly.	Reflection has four properties: (i) the image is the same size as the object; (ii) object and image are equidistant from the mirror line; (iii) the segment joining any object point to its corresponding image point is perpendicular to the mirror line — making the mirror line the perpendicular bisector of that segment; (iv) the image is laterally inverted relative to the object.	These four properties explain why the lorry in the matatu rear-view mirror appears as far behind the mirror glass as the actual lorry is in front of it (Property II — equidistance), why it appears the same size as the real lorry (Property I — equal size, confirmed by the tracing-paper fold showing congruence), and why the lorry's right headlight appears on the left in the mirror (Property IV — lateral inversion). The mirror line — the glass surface — is always the perpendicular bisector of every object-image connector, making the image's position completely predictable.
Lesson 5 Finding the Mirror Line — Determining the Equation of the Mirror Line Given an Object and Its Image	Learners observed that when point $P(2,1)$ and its image $P'(6,5)$ are plotted on a Cartesian plane, the segment PP' has a midpoint $M(4,3)$ and a gradient of 1. A line drawn through M with gradient -1 (the negative reciprocal) acts perfectly as the mirror line — every object point and its image are equidistant from this line and the connector is perpendicular to it.	The mirror line is always the perpendicular bisector of the segment joining an object point to its image. To find its equation: (1) calculate the midpoint M of the object-image segment; (2) find the gradient of the object-image segment and take its negative reciprocal to get the mirror line gradient; (3) substitute M and the gradient into $y - y_1 = m(x - x_1)$ to write the equation. This three-step algorithm works for any object-image pair, including cases where the mirror line is vertical ($x = k$) or horizontal ($y = k$).	The matatu conductor's mirror is positioned so that it is equidistant from every point of the lorry and the corresponding point of the lorry's image — it is the perpendicular bisector of every such pair. This is why the image appears exactly as far behind the mirror surface as the lorry is in front of it. Knowing how to find the mirror line equation algebraically confirms that the mirror's position is not arbitrary: it is the unique line that preserves all distances and angles, which is the geometric reason the image and object are the same size and shape — a foundation for proving congruence in the next lesson.
Lesson 6 Constructing Reflected Images — Drawing the Mirror Image of a Flag Outline on Plain Paper	Learners observed that when they dropped perpendiculars from each vertex of the Kenya flag polygon to the mirror line and marked equal distances on the other side, the resulting image vertices — when joined — formed a polygon that was the same	Every object point reflects to a unique image point located by the perpendicular bisector property (equal perpendicular distance on the opposite side of the mirror line); joining image points in order produces an image that is congruent to the object —	This explains why the matatu conductor's rear-view mirror shows the approaching lorry at exactly the same size as the real lorry — the mirror applies the perpendicular bisector property to every point on the lorry simultaneously, preserving all distances

	shape and size as the original, with every corresponding side length matching exactly in their measurement tables.	all lengths and angles are preserved regardless of the shape's complexity or the mirror line's orientation.	and angles. The image is not a shrunken or distorted copy; it is a congruent copy, which is why every measurable dimension of the lorry in the mirror matches the real lorry perfectly.
Lesson 7 Congruence Tests for Triangles — Proving the Matatu Mirror Image is Always an Exact Copy	Through compass-and-ruler construction tasks, learners observed that four specific sets of minimum information — three sides (SSS), two sides and an included angle (SAS), two angles and a side (AAS/ASA), and a right angle with hypotenuse and one side (RHS) — each produce exactly one unique triangle (up to orientation). They also observed that a reflected triangle always has the same three side lengths as its object, and that the reflection produces a mirror-flipped (opposite) copy.	Two triangles are congruent if they satisfy any one of four tests: SSS, SAS, AAS/ASA, or RHS. A reflected triangle is always congruent to its object by SSS (all three pairs of corresponding sides are equal, since reflection preserves all distances). This is always opposite (indirect) congruence because reflection reverses orientation. The four tests give the minimum conditions needed to guarantee congruence — it is not necessary to measure all six elements (three sides and three angles).	The matatu rear-view mirror phenomenon is now formally resolved: the image of the lorry in the mirror is congruent to the real lorry (by SSS) because reflection preserves all distances — every side of the lorry maps to an equal side in the image. The image is the same size and shape in every measurable dimension. The apparent 'reversal' (right headlight appearing on the left) is not a size or shape change — it is a flip of orientation, which is the defining property of opposite (indirect) congruence. This is why the lorry and its mirror image are always identical in every measurement, yet cannot be superimposed without flipping.
Lesson 8 Final Explanation — Unifying Reflection and Congruence to Solve the Matatu Mirror Mystery	Learners observed their own Lesson 1 prediction sketches alongside a completed Cartesian evidence pack showing the lorry ABCD reflected in the y-axis ($x = 0$), with all coordinates, side lengths, and the perpendicular bisector of one object–image segment labelled — confirming that the image is the same distance behind the mirror as the object is in front, that all side lengths are preserved, and that the image is laterally flipped.	Reflection maps every point according to a specific algebraic coordinate rule determined by the mirror line; the mirror line is always the perpendicular bisector of the segment joining any object point to its image; a reflected triangle is congruent to its object by SSS and the congruence is opposite (indirect) because the image cannot be overlaid on the object without lifting it — which is why the lorry's headlights appear on opposite sides in the mirror.	The matatu rear-view mirror acts like the y-axis of a coordinate plane: it maps every point on the approaching lorry to a unique image point directly opposite, at equal distance, according to the rule $(a, b) \rightarrow (-a, b)$. This preserves all lengths and angles, making the image congruent to the lorry by SSS. The lateral flip — the lorry's right headlight appearing on the left — is the signature of opposite (indirect) congruence: the image is a perfect mathematical copy, but it is a mirror copy, not a direct overlay, which is exactly what any flat reflective surface must produce.

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